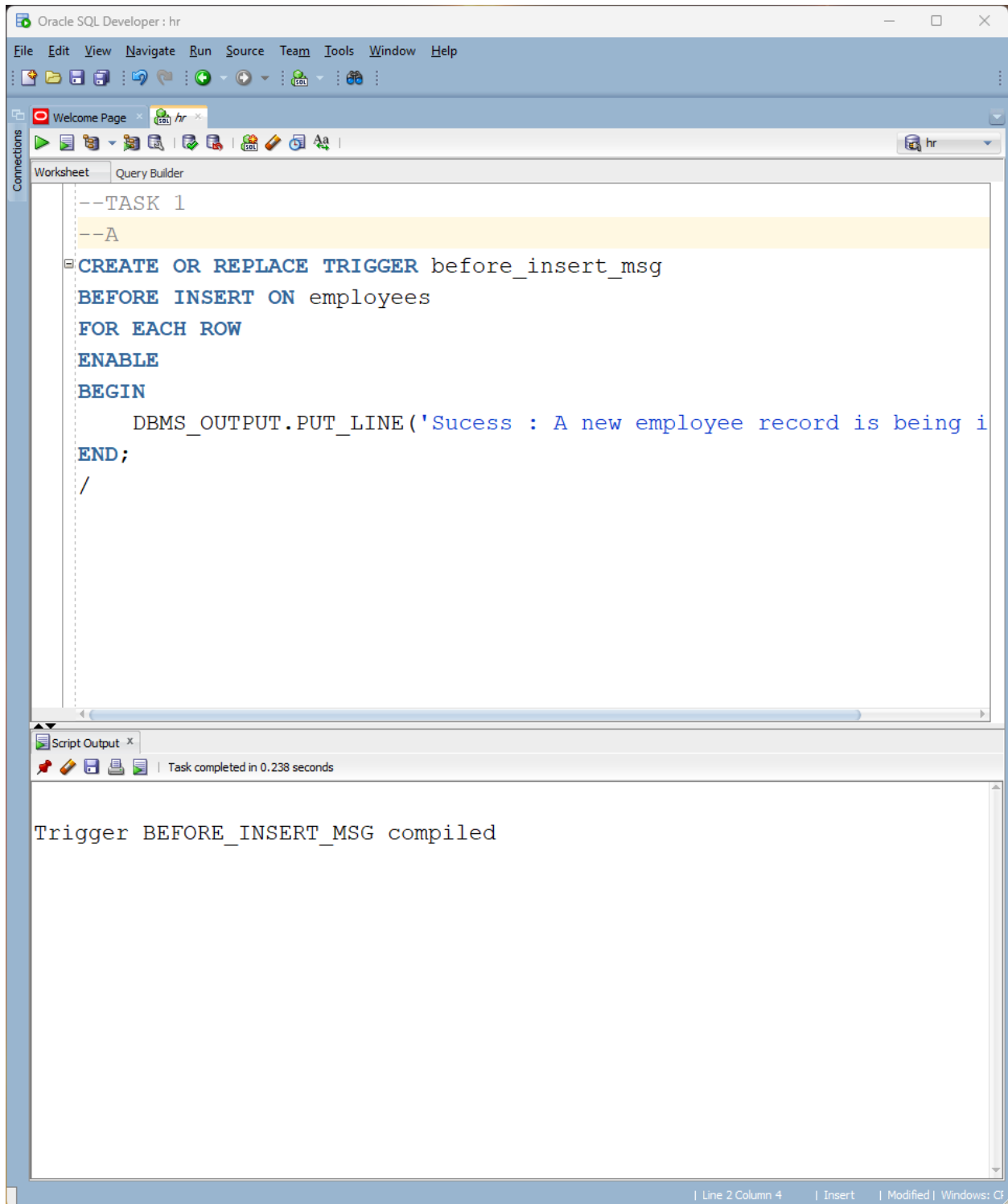
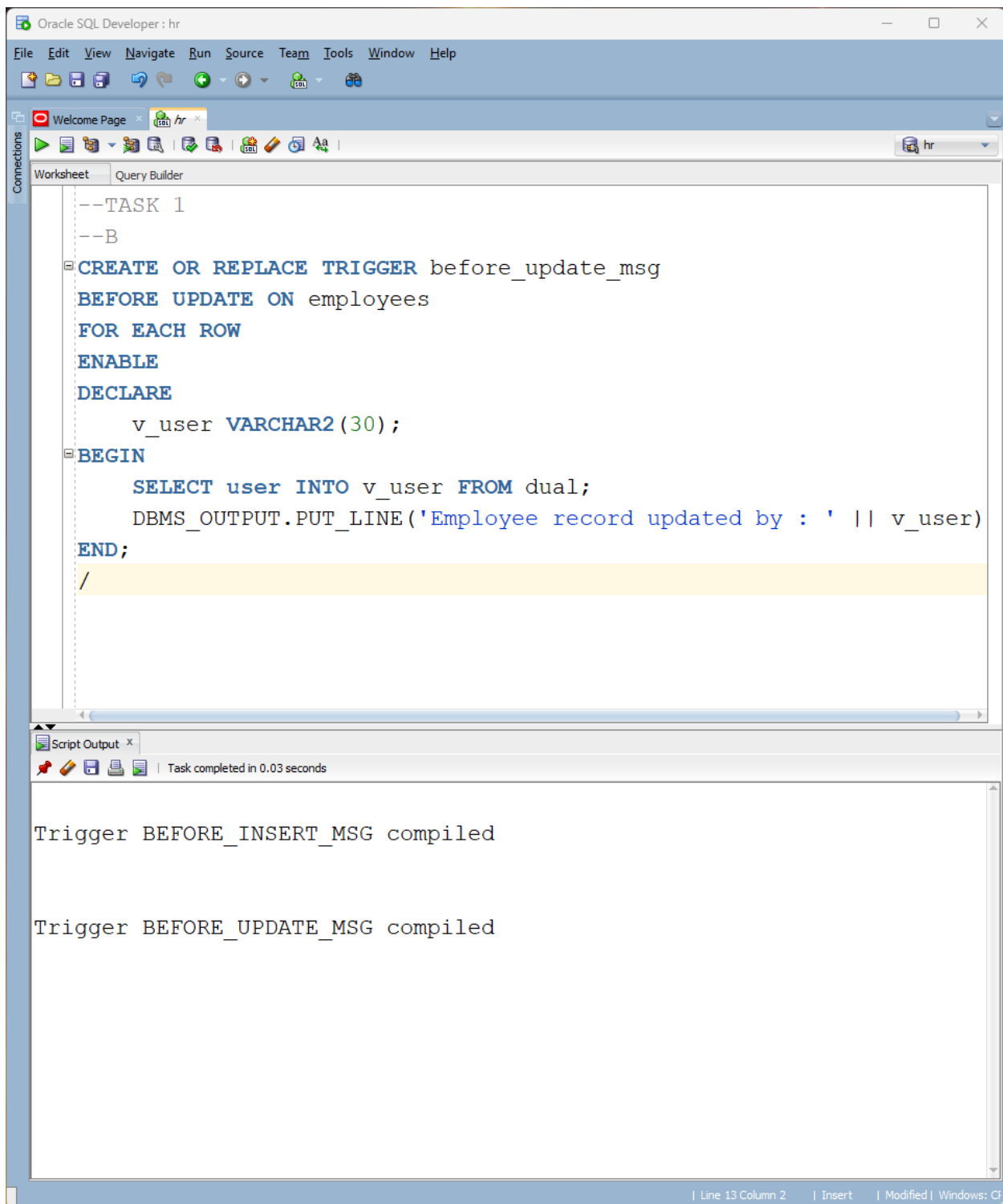
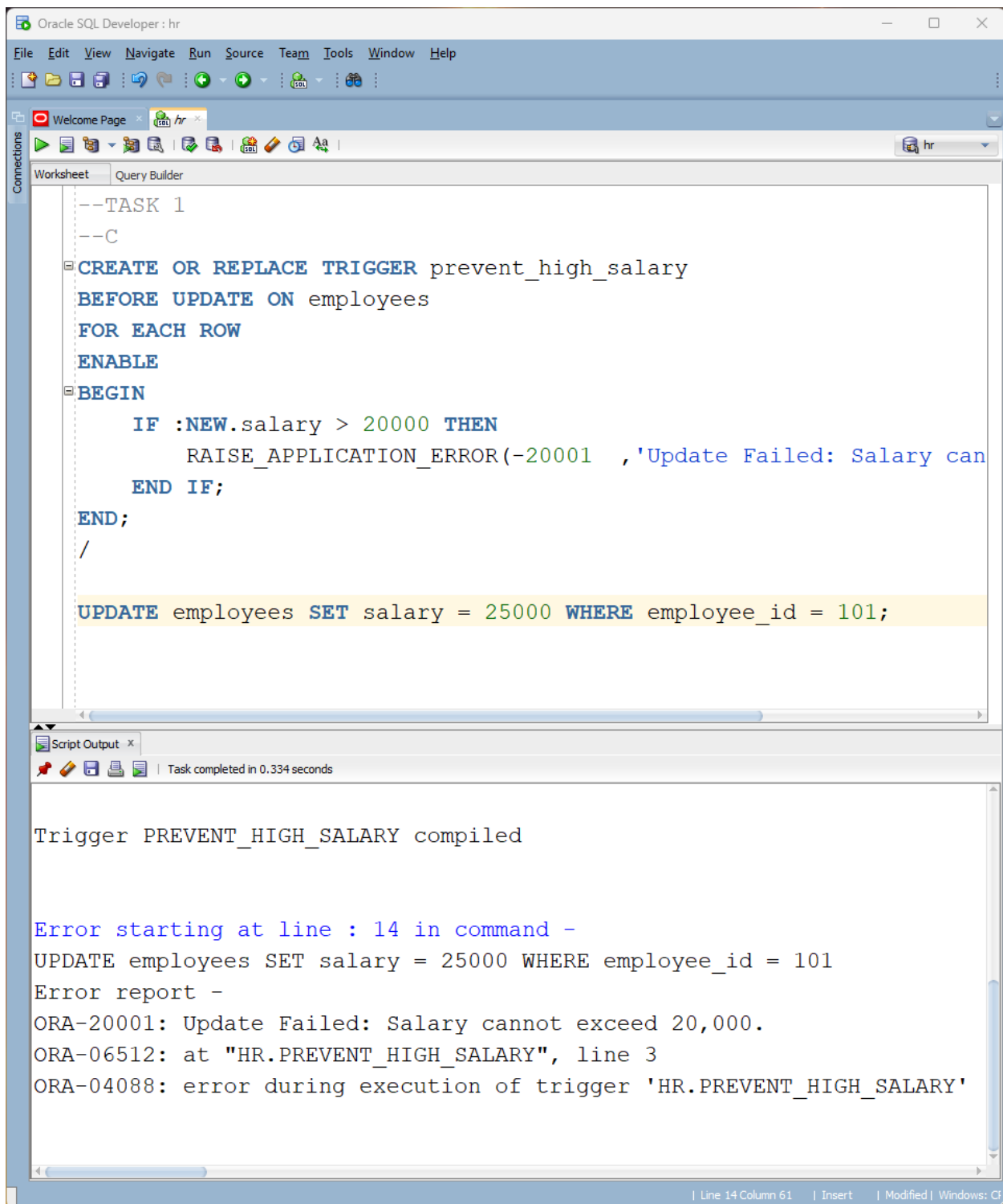
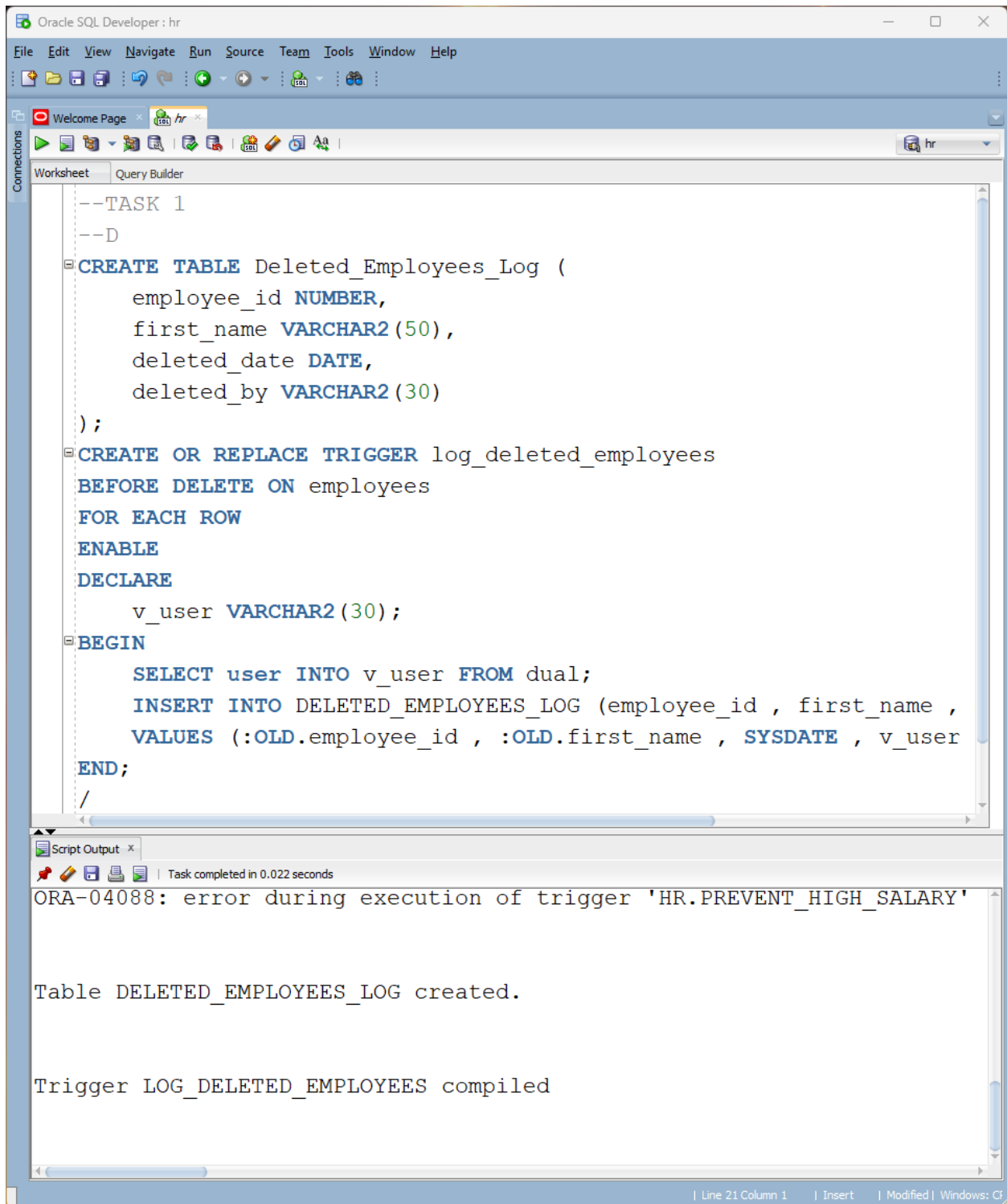


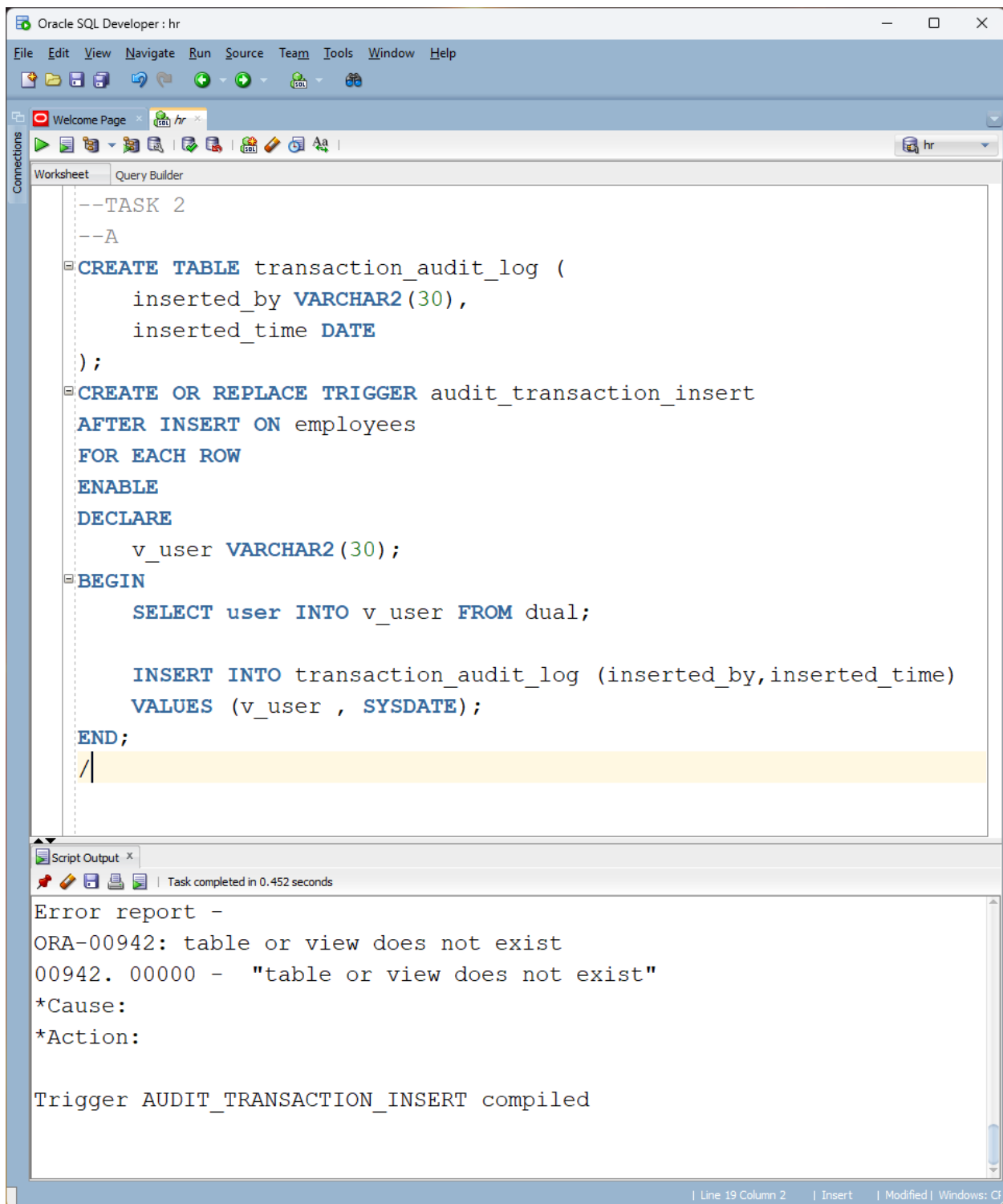
LAB 08

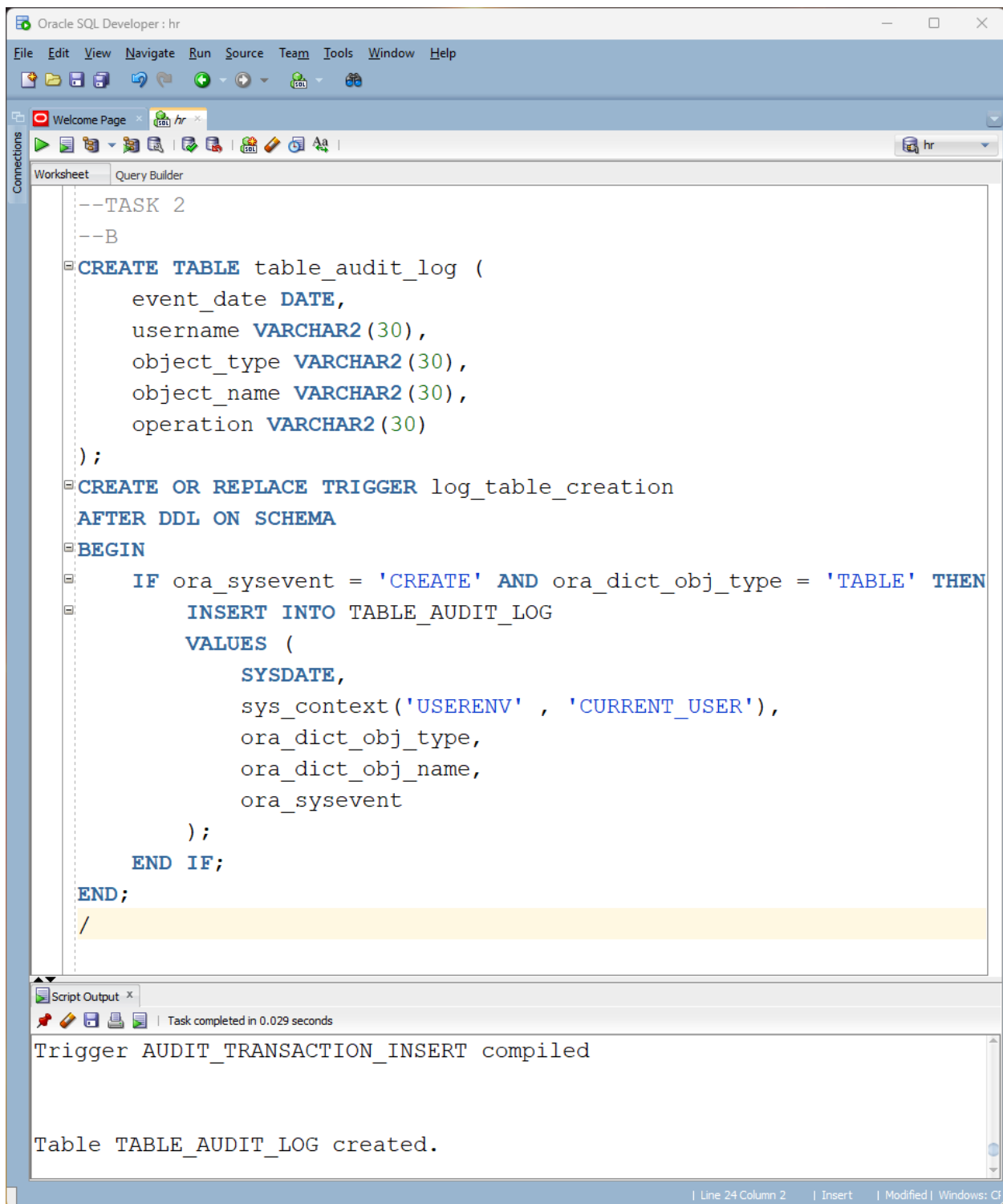


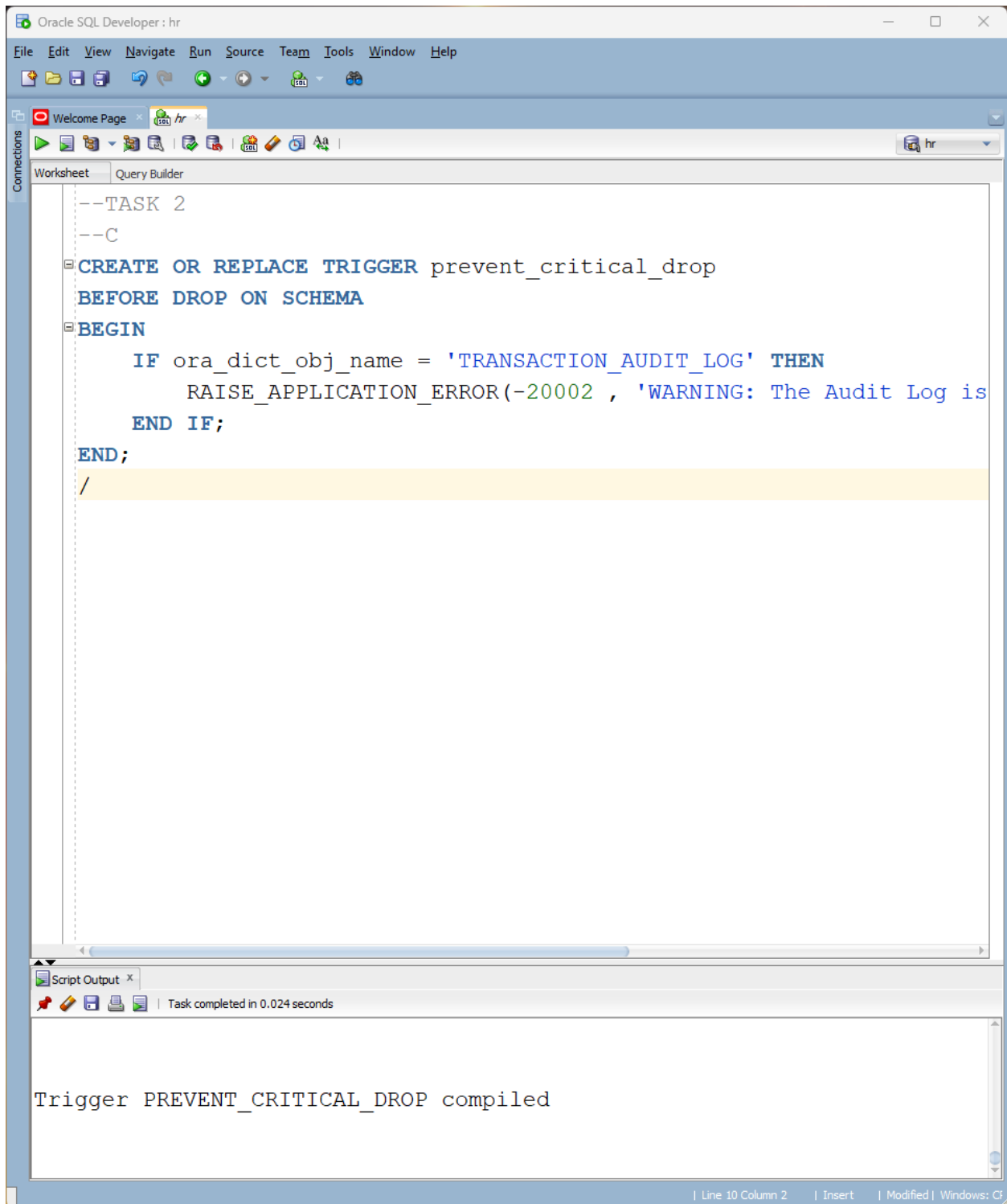


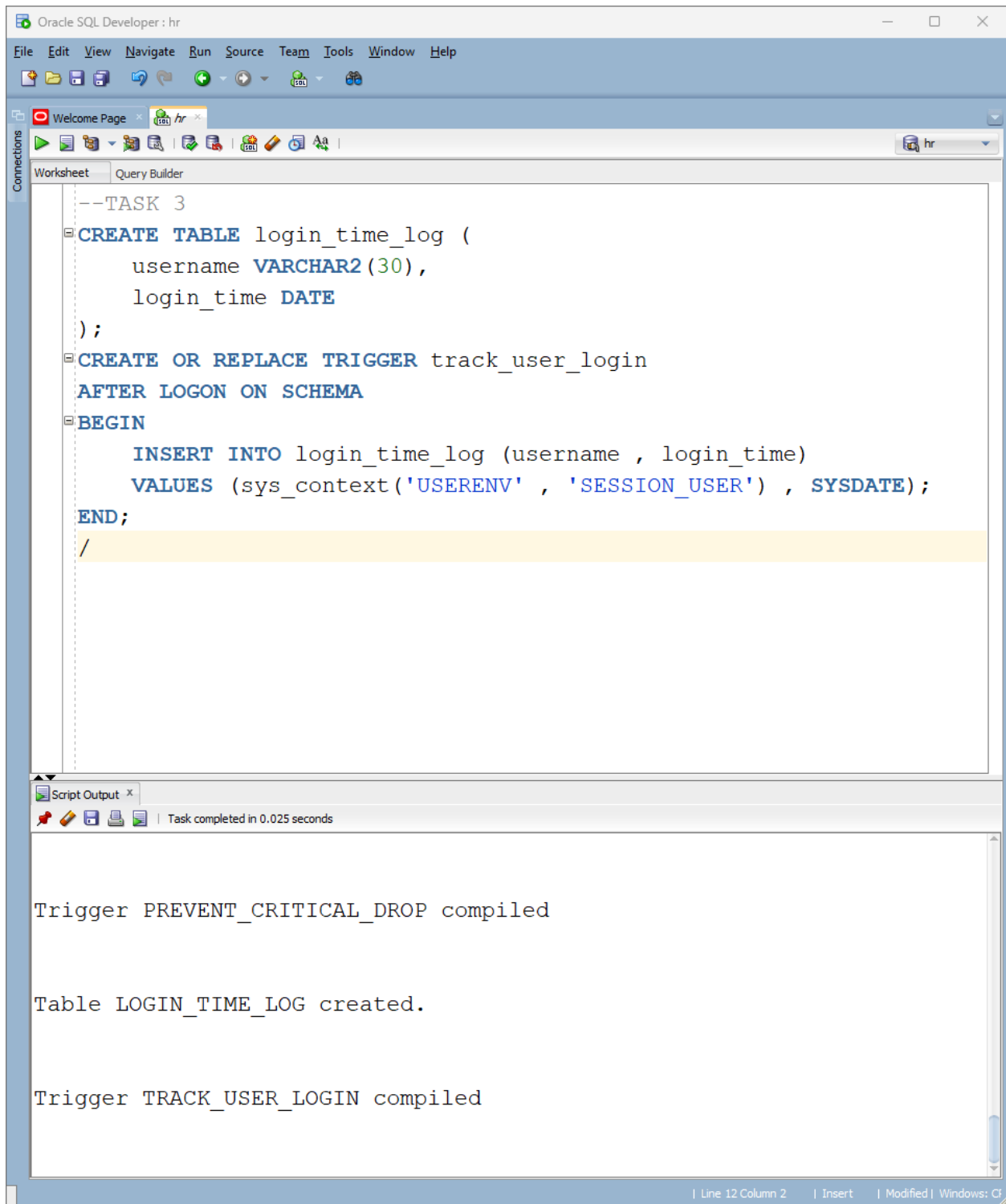


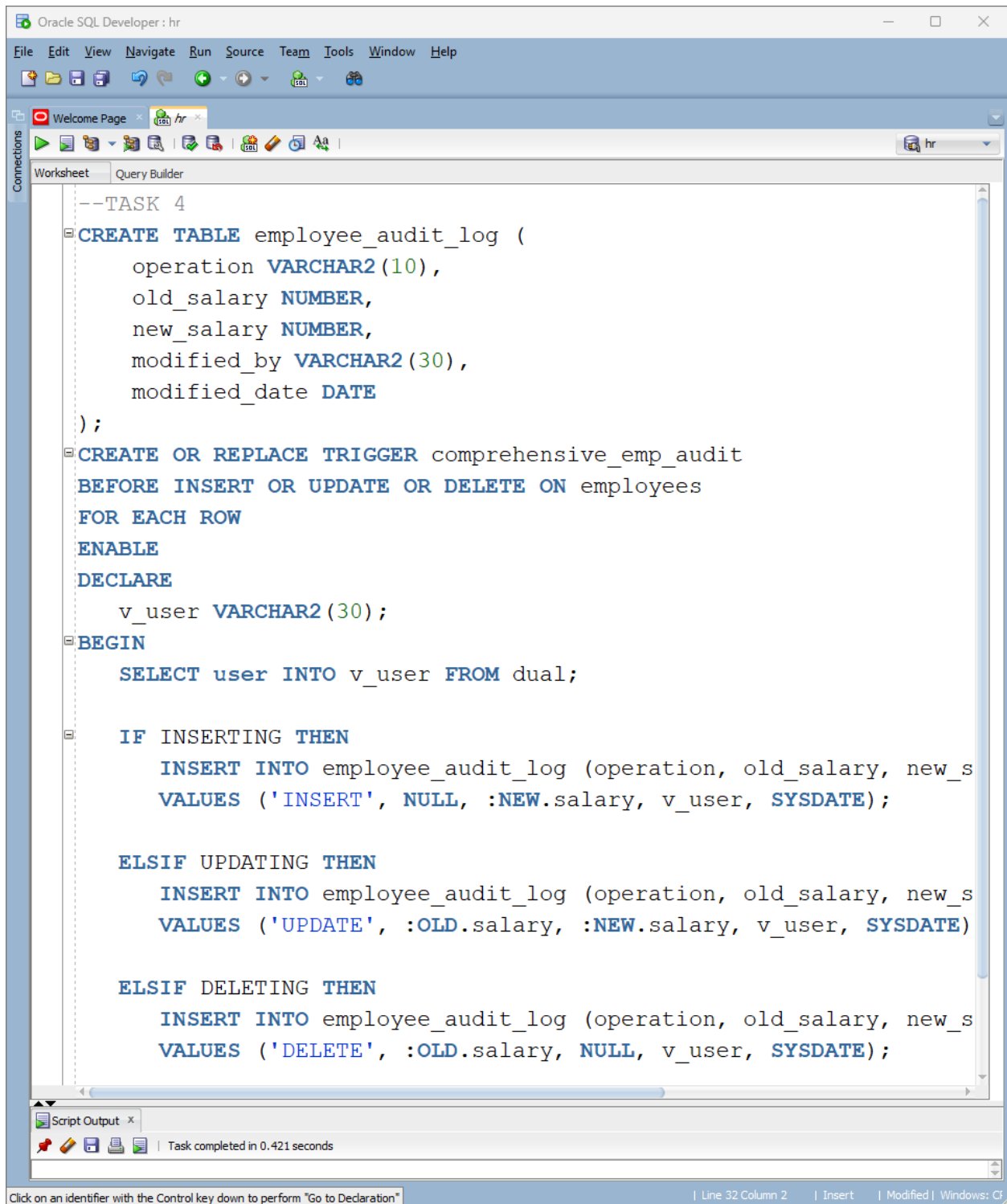


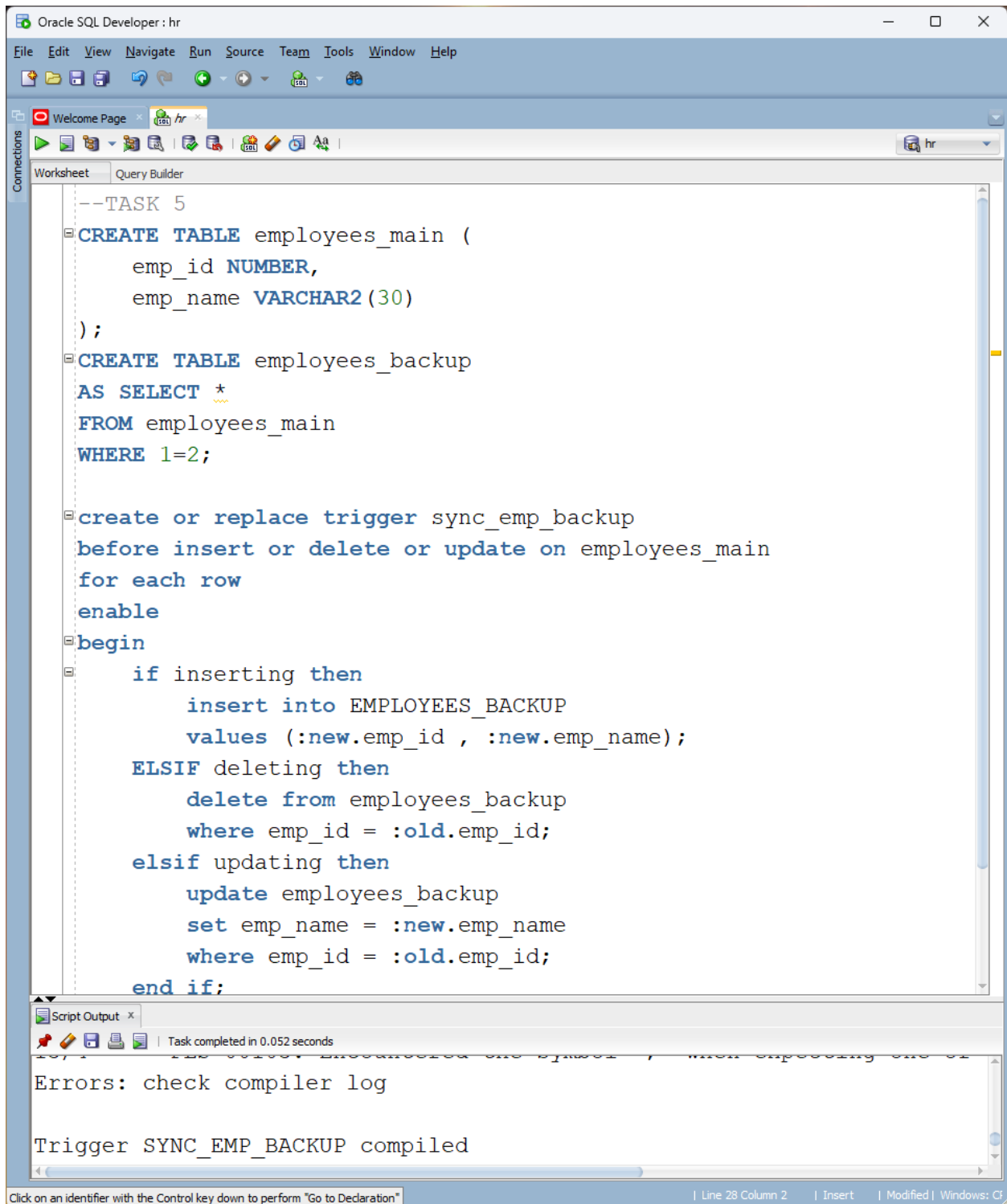


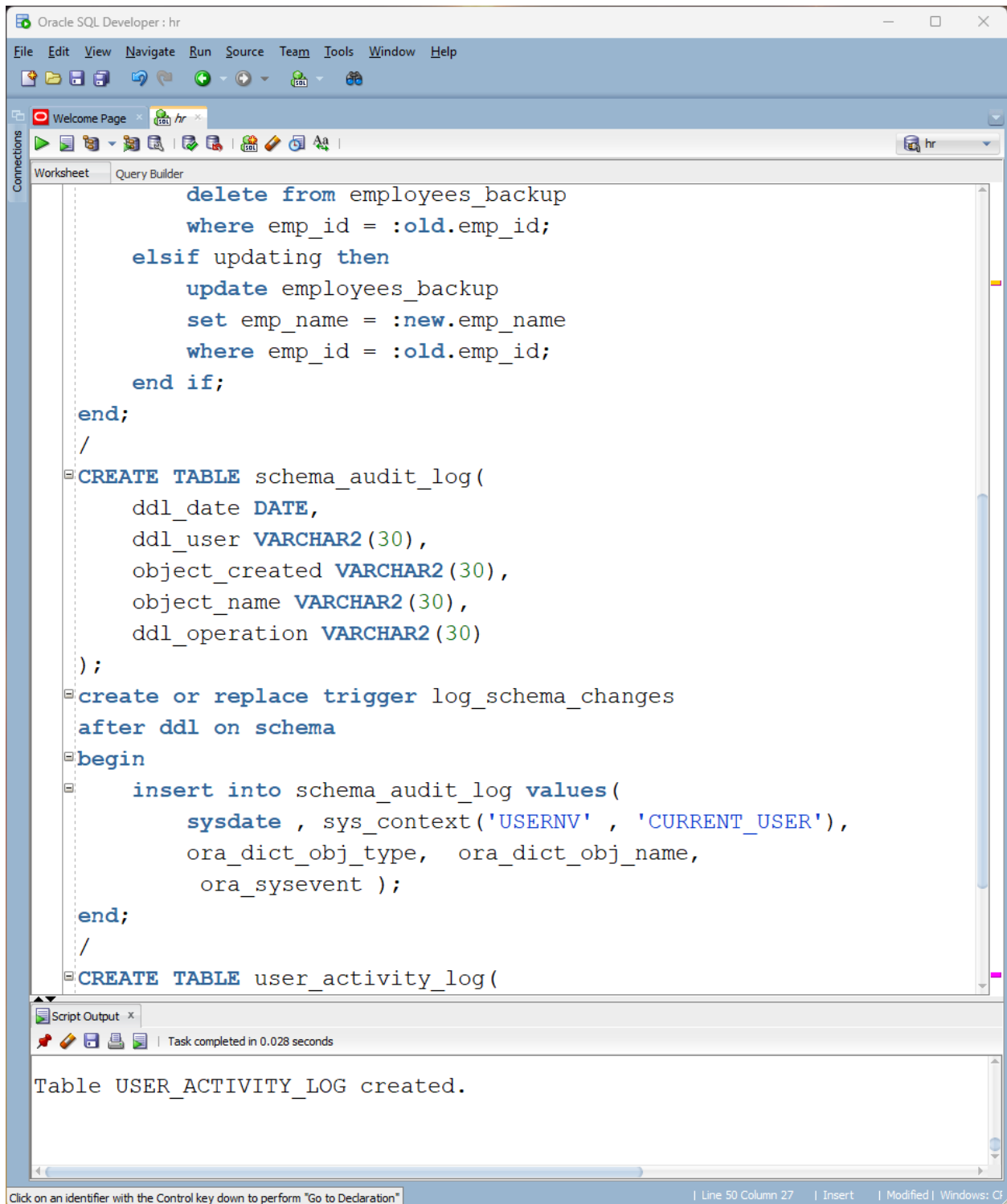


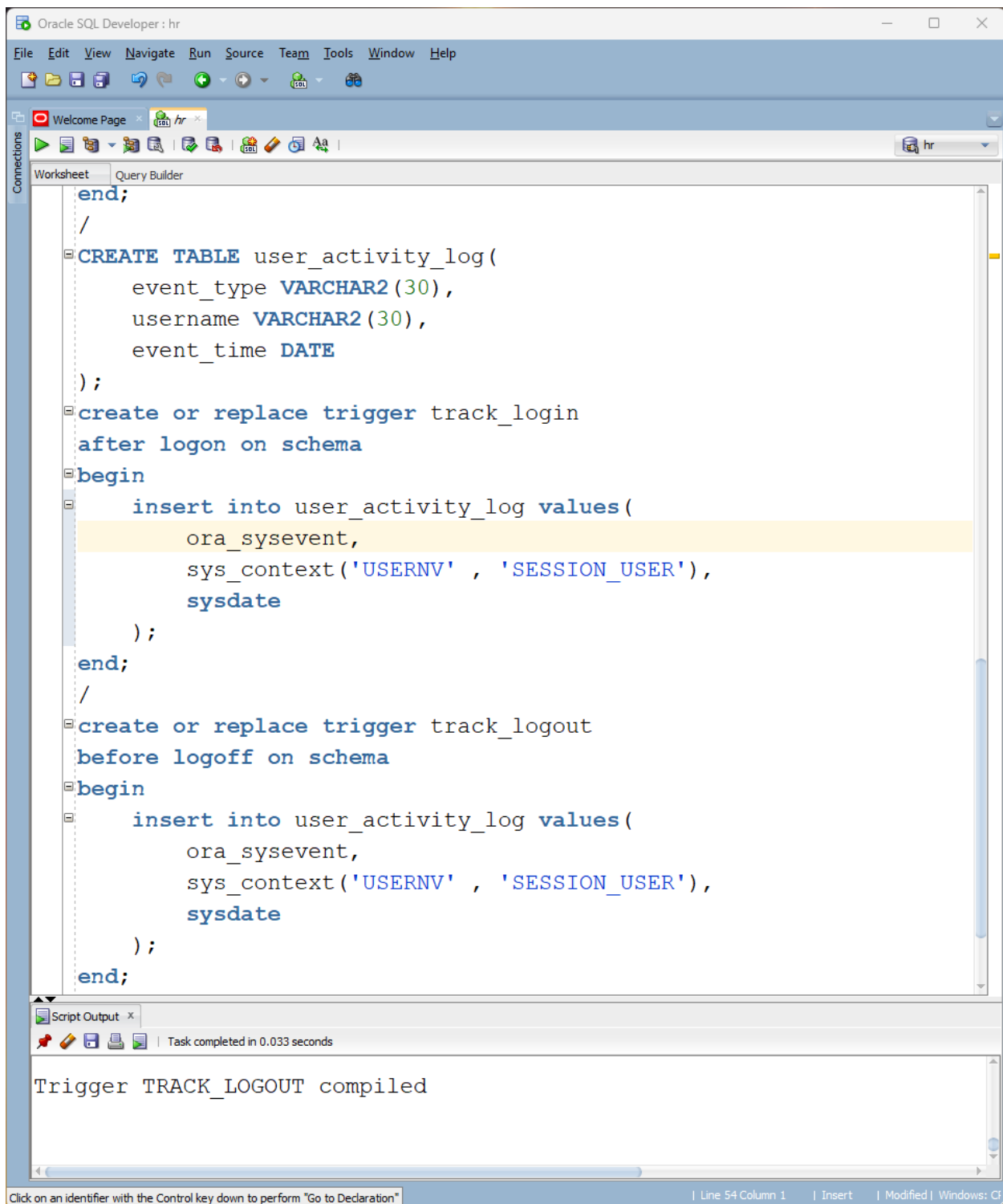


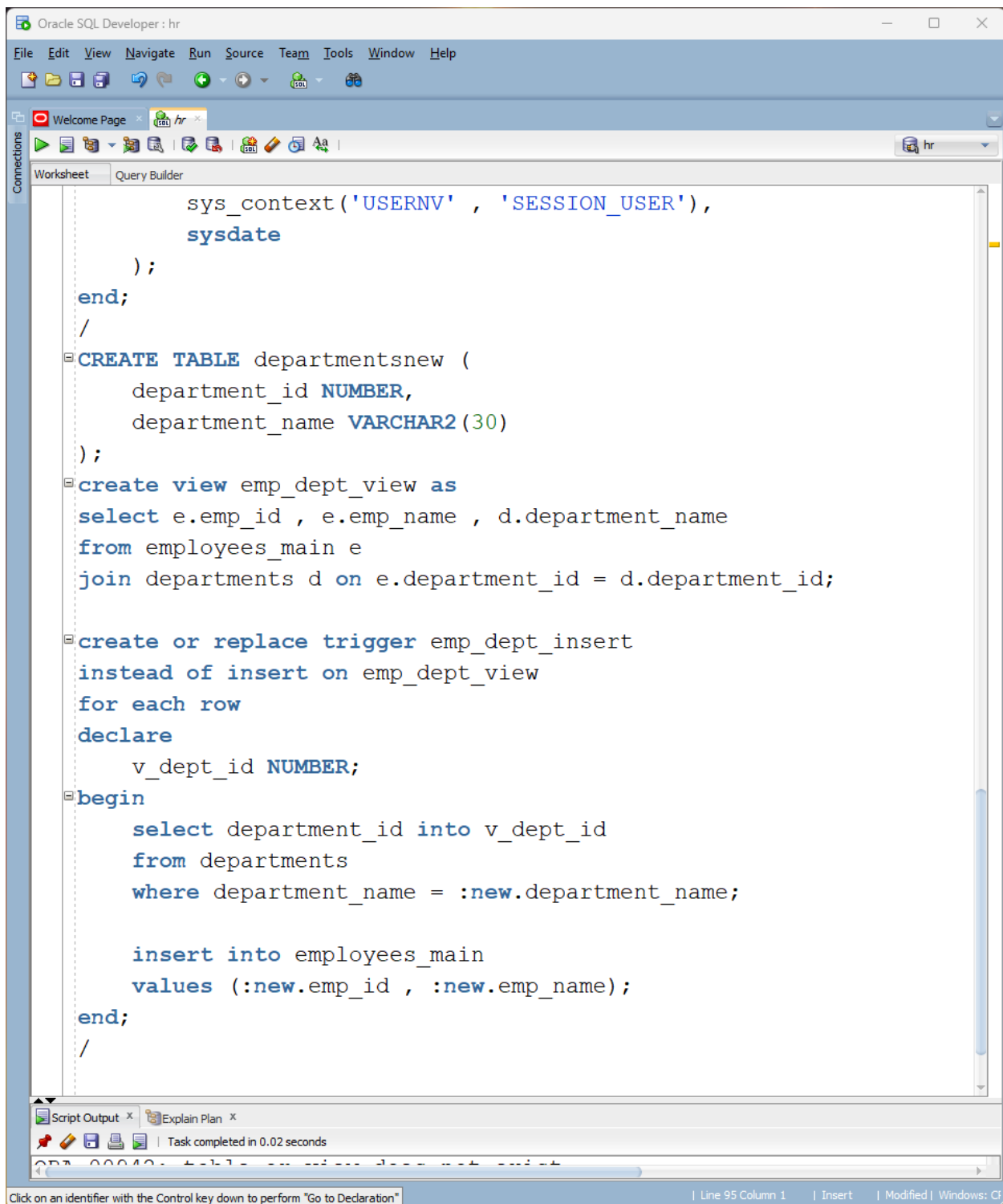












```
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File Edit View Navigate Run Source Tools Window Help
Welcome Page
Connections
Worksheet Query Builder

--TASK 6
CREATE TABLE enterprise_audit_log (
  operation_type VARCHAR2(10),
  old_employee_id NUMBER,
  new_employee_id NUMBER,
  old_salary NUMBER,
  new_salary NUMBER,
  performed_by VARCHAR2(30),
  action_timestamp DATE
);
create or replace trigger enterprise_audit_system
before insert or update or delete on employees
for each row
enable
declare
  v_user varchar(30);
begin
  select user into v_user from dual;

  if inserting then
    insert into enterprise_audit_log
    values ('INSERT', NULL, :NEW.employee_id, NULL, :NEW.salary, v_user, sysdate);
  elsif updating then
    insert into enterprise_audit_log
    values ('UPDATE', :OLD.employee_id, :NEW.employee_id, :OLD.salary, :NEW.salary, v_user, sysdate);
  elsif deleting then
    insert into enterprise_audit_log
    values ('DELETE', :OLD.employee_id, NULL, :OLD.salary, NULL, v_user, sysdate);
  end if;
end;

Script Output: X Explain Plan: X
Task completed in 0.051 seconds
Click on an identifier with the Control key down to perform "Go to Declaration" | Line 31 Column 2 | Smart | Modified | Windows: O
```

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File Edit View Navigate Run Source Tools Window Help
Welcome Page
Connections
Worksheet Query Builder

--TASK 7
-- 1. Create the main financial table
CREATE TABLE bank_accounts (
  account_id NUMBER,
  customer_name VARCHAR2(50),
  balance NUMBER
);

-- 2. Create the empty backup table using the 1=2 trick
CREATE TABLE bank_accounts_backup AS SELECT * FROM bank_accounts WHERE 1=2;

-- 3. Create the DML Trigger to sync them
CREATE OR REPLACE TRIGGER sync_bank_backup
BEFORE INSERT OR DELETE OR UPDATE ON bank_accounts
FOR EACH ROW
ENABLE
BEGIN
  IF inserting THEN
    INSERT INTO bank_accounts_backup (account_id, customer_name, balance)
    VALUES (:NEW.account_id, :NEW.customer_name, :NEW.balance);

  ELSIF deleting THEN
    DELETE FROM bank_accounts_backup
    WHERE account_id = :OLD.account_id;

  ELSIF updating THEN
    UPDATE bank_accounts_backup
    SET customer_name = :NEW.customer_name, balance = :NEW.balance
    WHERE account_id = :OLD.account_id;
end;

Script Output: X Explain Plan: X
Task completed in 0.172 seconds
ORA-00955: name is already used by an existing object
Click on an identifier with the Control key down to perform "Go to Declaration" | Line 115 Column 2 | Smart | Modified | Windows: O
Activate Windows
Go to Settings to activate Windows.
```

```
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File Edit View Navigate Run Source Team Tools Window Help
Welcome Page
Connections
Worksheet Query Builder
END IF;
END;
/
-- 1. Create the schema audit log table
CREATE TABLE financial_schema_audit (
    action_date DATE,
    action_user VARCHAR2(30),
    object_type VARCHAR2(30),
    object_name VARCHAR2(30),
    operation VARCHAR2(30)
);
-- 2. Create the DDL Trigger
CREATE OR REPLACE TRIGGER log_financial_schema
AFTER DDL ON SCHEMA
BEGIN
    INSERT INTO financial_schema_audit VALUES (
        SYSDATE,
        sys_context('USERENV', 'CURRENT_USER'),
        ora_dict_obj_type,
        ora_dict_obj_name,
        ora_sysevent
    );
END;
/
-- 1. Create the access log table
CREATE TABLE security_access_log (
    event_type VARCHAR2(30),
    username VARCHAR2(30),
    event_timestamp DATE
);
-- 2. Create the LOGON trigger
CREATE OR REPLACE TRIGGER bank_logon_audit
AFTER LOGON ON SCHEMA
BEGIN
    INSERT INTO security_access_log VALUES(
        ora_sysevent,
        sys_context('USERENV', 'SESSION_USER'),
        SYSDATE
    );
END;
/
-- 3. Create the LOGOFF trigger
CREATE OR REPLACE TRIGGER bank_logoff_audit
BEFORE LOGOFF ON SCHEMA
BEGIN
    INSERT INTO security_access_log VALUES(
        ora_sysevent,
        sys_context('USERENV', 'SESSION_USER'),
        SYSDATE
    );
END;
/
-- 1. Create a branches table
CREATE TABLE bank_branches (
    branch_id NUMBER
);
```

Script Output: Task completed in 0.172 seconds
ORA-00955: name is already used by an existing object
Click on an identifier with the Control key down to perform "Go to Declaration"

Activate Windows
Go to Settings to activate Windows.

Line 115 Column 2 | Smart | Modified | Windows: O

```
Oracle SQL Developer: hr
File Edit View Navigate Run Source Team Tools Window Help
Welcome Page
Connections
Worksheet Query Builder
username VARCHAR2(30),
event_timestamp DATE
);
-- 2. Create the LOGON trigger
CREATE OR REPLACE TRIGGER bank_logon_audit
AFTER LOGON ON SCHEMA
BEGIN
    INSERT INTO security_access_log VALUES(
        ora_sysevent,
        sys_context('USERENV', 'SESSION_USER'),
        SYSDATE
    );
END;
/
-- 3. Create the LOGOFF trigger
CREATE OR REPLACE TRIGGER bank_logoff_audit
BEFORE LOGOFF ON SCHEMA
BEGIN
    INSERT INTO security_access_log VALUES(
        ora_sysevent,
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    );
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/
-- 1. Create a branches table
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);
```

Script Output: Task completed in 0.172 seconds
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Line 115 Column 2 | Smart | Modified | Windows: O

